

WEEK 15: CRYPTOGRAPHY AND STEGANOGRAPHY

1. Android Pentesting

Android Penetration Testing (Android Pentesting) involves assessing Android applications and devices to identify security vulnerabilities that could be exploited by attackers. The goal is to find weaknesses in mobile applications, system configurations, and communication mechanisms and recommend measures to strengthen mobile security.

2. Cryptography

Cryptography is the practice of securing information by converting it into an unreadable format so that only authorized users can access it. It plays a vital role in protecting data confidentiality, integrity, authentication, and non-repudiation in digital communication and storage systems.

- Symmetric encryption (AES, DES)
- Asymmetric encryption (RSA)
- Hashing algorithms (MD5, SHA-1, SHA-256)
- Digital signatures and certificates
- Secure communication protocols

3. Steganography

Steganography is the technique of hiding secret information inside ordinary digital files such as images, audio, or video files in such a way that the existence of the hidden data is not noticeable. Unlike cryptography, which protects the content of information, steganography focuses on concealing the presence of the information itself.

Tools Used:

- Steghide: An open-source steganography tool used to embed and extract hidden data within image and audio files. It also supports encryption of the hidden data using a passphrase

Concepts Studied:

- Difference between steganography and cryptography
- Cover file, secret data, and stego file
- Embedding and extraction techniques
- Use of passphrases for added security
- Legitimate and malicious use cases of steganography